50 AWESOME SCIENCE EXPERIMENTS FO

Easy Projects You Can Do at Home (Ages 7-12)

Chemistry * Physics * Biology * Earth Science * Engineering

By Daniel Tesfamariam

50 AWESOME SCIENCE EXPERIMENTS FOR KIDS

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Published for Amazon KDP

Paperback Edition - 8.5 x 11 inches

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SAFETY FIRST!

1. Always have an adult nearby when doing experiments
2. Wear safety goggles if working with liquids
3. Never taste or eat experiment materials
4. Work in a well-ventilated area
5. Clean up your workspace when finished
6. Read ALL instructions before starting
7. If something goes wrong, tell an adult immediately
8. Wash your hands after every experiment

CHEMISTRY EXPERIMENTS

Experiments 1-10

EXPERIMENT #1

Baking Soda Volcano

Difficulty: Easy

Time: 10 min

MATERIALS:

* Baking soda
* Vinegar
* Food coloring
* Dish soap
* Container

STEPS:

1. Place container on a tray
2. Add 2 tbsp baking soda
3. Add a drop of dish soap
4. Add food coloring
5. Pour in vinegar and watch!

THE SCIENCE BEHIND IT:

The acetic acid in vinegar reacts with sodium bicarbonate to produce carbon dioxide gas. The bubbles create the eruption effect.

WHAT DID YOU OBSERVE?

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EXPERIMENT #2

Invisible Ink

Difficulty: Easy   Time: 15 min

MATERIALS:

* Lemon juice
* Cotton swab
* White paper
* Lamp or iron

STEPS:

1. Squeeze lemon juice into a cup
2. Dip cotton swab in juice
3. Write a secret message on paper
4. Let it dry completely
5. Hold paper near warm lamp to reveal

THE SCIENCE BEHIND IT:

Lemon juice is an organic substance that oxidizes when heated. The heat breaks down the compounds, turning them brown.

WHAT DID YOU OBSERVE?

EXPERIMENT #3

Color Changing Milk

Difficulty: Easy   Time: 10 min

MATERIALS:

* Whole milk
* Food coloring
* Dish soap
* Plate
* Cotton swab

STEPS:

1. Pour milk onto a plate
2. Add drops of different food colors
3. Dip cotton swab in dish soap
4. Touch the swab to the milk center
5. Watch colors dance and swirl!

THE SCIENCE BEHIND IT:

Dish soap breaks down fat molecules in milk. This creates movement that carries the food coloring in beautiful patterns.

WHAT DID YOU OBSERVE?

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EXPERIMENT #4

Homemade Slime

Difficulty: Medium

Time: 20 min

MATERIALS:

* White glue
* Borax
* Water
* Food coloring
* Bowl

STEPS:

1. Mix 1/2 cup glue with 1/2 cup water
2. Add food coloring and stir
3. In another cup mix 1 tsp borax with 1 cup warm water
4. Slowly add borax solution to glue mixture
5. Knead until slime forms

THE SCIENCE BEHIND IT:

Borax creates cross-links between the polymer chains in glue. This turns the liquid glue into a stretchy solid-like substance.

WHAT DID YOU OBSERVE?

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EXPERIMENT #5

Crystal Growing

Difficulty: Medium Time: 3 days

MATERIALS:

* Sugar or salt
* Hot water
* Jar
* String
* Pencil

STEPS:

1. Heat water until very hot (adult help)
2. Stir in sugar until no more dissolves
3. Pour solution into jar
4. Hang string from pencil across jar top
5. Wait 3-7 days for crystals to form

THE SCIENCE BEHIND IT:

As water evaporates, dissolved molecules come together in an orderly pattern. This repeated pattern creates the geometric crystal shape.

WHAT DID YOU OBSERVE?

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EXPERIMENT #6

Density Tower

Difficulty: Easy   Time: 15 min

MATERIALS:

* Honey
* Corn syrup
* Dish soap
* Water
* Vegetable oil
* Rubbing alcohol
* Tall glass

STEPS:

1. Pour honey into glass first
2. Slowly add corn syrup
3. Add dish soap carefully
4. Pour water very slowly
5. Add vegetable oil
6. Top with rubbing alcohol

THE SCIENCE BEHIND IT:

Each liquid has a different density (weight per volume). Heavier liquids sink below lighter ones, creating distinct layers.

WHAT DID YOU OBSERVE?

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EXPERIMENT #7

pH Testing with Cabbage

Difficulty: Medium   Time: 25 min

MATERIALS:

* Red cabbage
* Hot water
* Clear cups
* Lemon juice
* Baking soda
* Vinegar
* Soap

STEPS:

1. Chop cabbage and soak in hot water 10 min
2. Strain purple liquid into cups
3. Add different household liquids to each cup
4. Observe color changes
5. Record which are acids and which are bases

THE SCIENCE BEHIND IT:

Red cabbage contains anthocyanin, a natural pH indicator. It turns red in acids, purple in neutral, and green-blue in bases.

WHAT DID YOU OBSERVE?

EXPERIMENT #8

Elephant Toothpaste

Difficulty: Medium

Time: 15 min

MATERIALS:

* Hydrogen peroxide 3%
* Yeast
* Warm water
* Dish soap
* Food coloring
* Bottle

STEPS:

1. Pour hydrogen peroxide into bottle
2. Add food coloring and dish soap
3. Mix yeast with warm water in a cup
4. Pour yeast mixture into bottle quickly
5. Stand back and watch the foam!

THE SCIENCE BEHIND IT:

Yeast contains catalase enzyme that rapidly breaks down hydrogen peroxide into water and oxygen gas. The soap traps oxygen in foam.

WHAT DID YOU OBSERVE?

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EXPERIMENT #9

Penny Cleaning

Difficulty: Easy   Time: 10 min

MATERIALS:

* Dirty pennies
* Vinegar
* Salt
* Cup
* Paper towels

STEPS:

1. Mix 1/4 cup vinegar with 1 tsp salt
2. Drop dirty pennies into solution
3. Wait 30 seconds
4. Remove and rinse with water
5. Compare to uncleaned pennies

THE SCIENCE BEHIND IT:

The acid in vinegar dissolves copper oxide (the dark coating). Salt makes the acid stronger. The result is shiny clean copper.

WHAT DID YOU OBSERVE?

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EXPERIMENT #10

CO2 Balloon Inflator

Difficulty: Easy Time: 10 min

MATERIALS:

* Balloon
* Baking soda
* Vinegar
* Empty bottle
* Funnel

STEPS:

1. Pour vinegar into the bottle (1/3 full)
2. Use funnel to put baking soda in balloon
3. Stretch balloon opening over bottle top
4. Lift balloon so soda falls into vinegar
5. Watch balloon inflate with CO2 gas

THE SCIENCE BEHIND IT:

Baking soda and vinegar react to produce carbon dioxide gas. The gas expands and fills the balloon, inflating it without blowing.

WHAT DID YOU OBSERVE?

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PHYSICS EXPERIMENTS

Experiments 11-20

EXPERIMENT #11

Static Electricity Butterfly

Difficulty: Easy　　Time: 10 min

MATERIALS:

* Tissue paper
* Balloon
* Scissors
* Wool cloth

STEPS:

1. Cut tissue paper into butterfly shape
2. Blow up balloon and tie it
3. Rub balloon vigorously on wool
4. Hold balloon near tissue butterfly
5. Watch it fly up to the balloon!

THE SCIENCE BEHIND IT:

Rubbing creates static electricity by transferring electrons. The charged balloon attracts the neutral tissue paper through electrostatic force.

WHAT DID YOU OBSERVE?

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EXPERIMENT #12

Egg Drop Challenge

Difficulty: Hard   Time: 30 min

MATERIALS:

* Raw egg
* Straws
* Tape
* Cotton balls
* Plastic bags
* Newspaper

STEPS:

1. Design a container to protect the egg
2. Use materials to cushion and slow the fall
3. Build your contraption around the egg
4. Drop from increasing heights
5. Redesign if egg breaks!

THE SCIENCE BEHIND IT:

The goal is to reduce impact force by increasing deceleration time.
Soft materials absorb energy and distribute force over more area.

WHAT DID YOU OBSERVE?

EXPERIMENT #13

Bernoulli Ball Float

Difficulty: Easy

Time: 5 min

MATERIALS:

* Hair dryer
* Ping pong ball
* Funnel optional

STEPS:

1. Turn hair dryer on high pointing up
2. Place ping pong ball in air stream
3. Let go - it floats!
4. Tilt dryer slowly - ball stays
5. Try with different sized balls

THE SCIENCE BEHIND IT:

Fast-moving air has lower pressure than still air (Bernoulli's principle). The ball stays centered because surrounding higher pressure pushes it back.

WHAT DID YOU OBSERVE?

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EXPERIMENT #14

Magnet Exploration

Difficulty: Easy Time: 20 min

MATERIALS:

* Bar magnet
* Various objects (metal/non-metal)
* Paper clips
* Paper
* Compass

STEPS:

1. Test which objects are attracted to magnet
2. Try magnet through paper and cardboard
3. Make a chain of paper clips
4. Observe compass needle near magnet
5. Map the magnetic field with iron filings

THE SCIENCE BEHIND IT:

Magnets attract ferromagnetic materials (iron, nickel, cobalt).
Magnetic fields can pass through non-magnetic materials like paper.

WHAT DID YOU OBSERVE?

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EXPERIMENT #15

Simple Catapult

Difficulty: Medium   Time: 20 min

MATERIALS:

* Popsicle sticks (10)
* Rubber bands
* Plastic spoon
* Pompom or marshmallow

STEPS:

1. Stack 8 sticks and rubber band both ends
2. Place 1 stick perpendicular on top
3. Rubber band the crossing point
4. Attach spoon to top stick
5. Load pompom and launch!

THE SCIENCE BEHIND IT:

The bent stick stores elastic potential energy. When released, it converts to kinetic energy, launching the projectile in an arc.

WHAT DID YOU OBSERVE?

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EXPERIMENT #16

Light Refraction Rainbow

Difficulty: Easy Time: 10 min

MATERIALS:

* Glass of water
* White paper
* Sunlight or flashlight
* Mirror optional

STEPS:

1. Fill glass with water completely
2. Place on edge of table in sunlight
3. Position white paper on floor below
4. Adjust until rainbow appears on paper
5. Identify all 7 colors of the spectrum

THE SCIENCE BEHIND IT:

White light contains all colors. Water bends (refracts) each color at slightly different angles, separating them into a visible spectrum.

WHAT DID YOU OBSERVE?

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EXPERIMENT #17

Pendulum Waves

Difficulty: Medium Time: 25 min

MATERIALS:

* String (different lengths)
* Nuts or washers
* Ruler or dowel
* Tape
* Stopwatch

STEPS:

1. Cut 5 strings of slightly different lengths
2. Tie a nut to each string end
3. Attach all strings to a ruler
4. Pull all back equally and release together
5. Watch the mesmerizing wave patterns

THE SCIENCE BEHIND IT:

Pendulum period depends on length. Different lengths swing at different rates, creating beautiful wave patterns that shift over time.

WHAT DID YOU OBSERVE?

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EXPERIMENT #18

Sound Vibration Viewer

Difficulty: Easy   Time: 10 min

MATERIALS:

* Plastic wrap
* Bowl or cup
* Rice or salt grains
* Speaker or pan

STEPS:

1. Stretch plastic wrap tightly over bowl
2. Place rice grains on top
3. Hold near speaker playing music
4. Watch grains dance to the beat
5. Try different volumes and pitches

THE SCIENCE BEHIND IT:

Sound is vibration traveling through air. These vibrations hit the plastic wrap, making it vibrate too, which bounces the rice grains.

WHAT DID YOU OBSERVE?

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EXPERIMENT #19

Inertia Tablecloth Trick

Difficulty: Medium		Time: 15 min

MATERIALS:

* Smooth cloth
* Plastic plate
* Plastic cup (empty)
* Smooth table

STEPS:

1. Place cloth on smooth table edge
2. Put plastic plate on cloth
3. Place empty cup on plate
4. Grab cloth edge firmly
5. Pull straight down FAST (not out)

THE SCIENCE BEHIND IT:

Objects at rest tend to stay at rest (Newton’s First Law). Pulling fast means friction doesn’t have time to transfer enough force to move dishes.

WHAT DID YOU OBSERVE?

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EXPERIMENT #20

Paper Airplane Science

Difficulty: Easy

Time: 20 min

MATERIALS:

* Paper (various types)
* Tape measure
* Timer
* Notebook

STEPS:

1. Fold 5 different airplane designs
2. Throw each 3 times and measure distance
3. Record which flies farthest
4. Modify wings and test again
5. Find what makes the best flyer

THE SCIENCE BEHIND IT:

Four forces act on planes: thrust, drag, lift, and gravity. Wing shape and angle determine how air flows, affecting lift and distance.

WHAT DID YOU OBSERVE?

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BIOLOGY EXPERIMENTS

Experiments 21-30

EXPERIMENT #21

Seed Germination Lab

Difficulty: Easy   Time: 7 days

MATERIALS:

* Bean seeds
* Paper towels
* Plastic bags
* Water
* Tape

STEPS:

1. Wet paper towel (damp not soaking)
2. Place 3 seeds on towel
3. Put towel in plastic bag (leave open)
4. Tape to sunny window
5. Record growth daily for 1 week

THE SCIENCE BEHIND IT:

Seeds contain an embryo plant and food supply. Water activates enzymes that break down stored food, giving energy for the sprout to grow.

WHAT DID YOU OBSERVE?

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EXPERIMENT #22

Build a Terrarium

Difficulty: Medium Time: 30 min

MATERIALS:

* Clear jar or bottle
* Small rocks
* Charcoal
* Potting soil
* Small plants
* Water spray

STEPS:

1. Layer rocks at bottom (1 inch)
2. Add thin charcoal layer
3. Add 2-3 inches of soil
4. Plant small plants carefully
5. Spray with water and seal

THE SCIENCE BEHIND IT:

Terrariums are miniature ecosystems. Water evaporates, condenses on walls, and falls back like rain. Plants produce oxygen and recycle CO_2.

WHAT DID YOU OBSERVE?

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EXPERIMENT #23

Celery Color Transport

Difficulty: Easy   Time: 24 hours

MATERIALS:

* Celery stalks with leaves
* Food coloring
* Clear glasses
* Water
* Knife (adult help)

STEPS:

1. Fill glasses with water and add food color
2. Cut celery bottom at angle
3. Place stalks in colored water
4. Wait 24 hours
5. Cut stalk open to see colored tubes

THE SCIENCE BEHIND IT:

Plants transport water through xylem tubes (like tiny straws).
Capillary action and transpiration pull water up from roots to leaves.

WHAT DID YOU OBSERVE?

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EXPERIMENT #24

Yeast Alive Experiment

Difficulty: Easy Time: 15 min

MATERIALS:

* Active dry yeast
* Warm water
* Sugar
* Balloon
* Empty bottle

STEPS:

1. Add warm water to bottle
2. Stir in 1 packet of yeast
3. Add 1 tbsp sugar
4. Stretch balloon over bottle top
5. Watch balloon inflate over 15 minutes

THE SCIENCE BEHIND IT:

Yeast are living single-celled fungi. They eat sugar and produce carbon dioxide gas and alcohol through fermentation. The gas inflates the balloon.

WHAT DID YOU OBSERVE?

EXPERIMENT #25

Fingerprint Detective

Difficulty: Easy Time: 20 min

MATERIALS:

* Inkpad or pencil graphite
* Clear tape
* White paper
* Magnifying glass
* Index cards

STEPS:

1. Press finger on inkpad or penciled paper
2. Press finger onto white paper clearly
3. Use magnifying glass to examine patterns
4. Compare prints from all 10 fingers
5. Classify as loop, whorl, or arch

THE SCIENCE BEHIND IT:

Fingerprints form before birth and never change. The three main patterns are loops (60%), whorls (35%), and arches (5%). No two are identical.

WHAT DID YOU OBSERVE?

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EXPERIMENT #26

Owl Pellet Dissection

Difficulty: Medium Time: 30 min

MATERIALS:

* Owl pellet (buy online)
* Toothpicks
* Tweezers
* Paper plate
* Bone chart

STEPS:

1. Soak pellet in water 10 minutes
2. Gently pull apart with toothpicks
3. Separate fur/feathers from bones
4. Identify bones using chart
5. Reconstruct the prey skeleton

THE SCIENCE BEHIND IT:

Owls swallow prey whole. They cannot digest bones, fur, and feathers. These form a pellet that the owl regurgitates. It reveals what they ate.

WHAT DID YOU OBSERVE?

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EXPERIMENT #27

Mold Bread Experiment

Difficulty: Easy

Time: 7 days

MATERIALS:

* Bread slices (no preservatives)
* Plastic bags
* Water
* Labels
* Dark place

STEPS:

1. Put dry bread in one sealed bag
2. Put damp bread in another sealed bag
3. Label and put both in dark warm spot
4. Check daily without opening
5. Record mold growth differences

THE SCIENCE BEHIND IT:

Mold is a fungus that needs moisture, warmth, food, and darkness to grow. The wet bread provides ideal conditions for rapid mold colonization.

WHAT DID YOU OBSERVE?

EXPERIMENT #28

Heart Rate Explorer

Difficulty: Easy

Time: 15 min

MATERIALS:

* Stopwatch or timer
* Notebook
* Pencil
* Jump rope optional

STEPS:

1. Find your pulse (wrist or neck)
2. Count beats for 15 sec, multiply by 4
3. Record resting heart rate
4. Do jumping jacks for 1 minute
5. Immediately measure again and compare

THE SCIENCE BEHIND IT:

Your heart pumps faster during exercise to deliver more oxygen to working muscles. Fit people’s hearts recover to resting rate faster.

WHAT DID YOU OBSERVE?

EXPERIMENT #29

Photosynthesis Test

Difficulty: Medium Time: 3 days

MATERIALS:

* Plant with large leaves
* Aluminum foil
* Tape
* Sunlight

STEPS:

1. Cover half of one leaf with foil
2. Tape foil securely in place
3. Leave plant in sunlight for 3 days
4. Remove foil and compare leaf halves
5. Covered half will be lighter/yellow

THE SCIENCE BEHIND IT:

Plants need light to make food (photosynthesis). Without light, chlorophyll breaks down and the leaf cannot produce glucose for energy.

WHAT DID YOU OBSERVE?

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EXPERIMENT #30

DNA Extraction

Difficulty: Medium   Time: 20 min

MATERIALS:

* Strawberry
* Dish soap
* Salt
* Rubbing alcohol (cold)
* Plastic bag
* Coffee filter
* Cup

STEPS:

1. Mash strawberry in sealed bag
2. Add soap and salt solution
3. Filter through coffee filter into cup
4. Slowly pour cold alcohol on top
5. Watch white stringy DNA appear!

THE SCIENCE BEHIND IT:

Soap breaks open cell membranes. Salt clumps proteins. DNA is less dense than alcohol, so it rises and becomes visible as white strands.

WHAT DID YOU OBSERVE?

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EARTH SCIENCE EXPERIMENTS

Experiments 31-40

EXPERIMENT #31

Rock Cycle in a Cup

Difficulty: Easy　　　　Time: 20 min

MATERIALS:

* Crayon shavings (3 colors)
* Aluminum foil
* Warm water
* Cup
* Wax paper

STEPS:

1. Shave crayons into small pieces by color
2. Layer shavings in foil cup (sediment)
3. Press hard with thumb (sedimentary rock)
4. Place in warm water to soften (metamorphic)
5. Melt fully and cool (igneous rock)

THE SCIENCE BEHIND IT:

The rock cycle shows how rocks transform. Pressure creates sedimentary, heat and pressure make metamorphic, and melting/cooling creates igneous.

WHAT DID YOU OBSERVE?

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EXPERIMENT #32

Earthquake Simulator

Difficulty: Easy Time: 15 min

MATERIALS:

* Jello (made and set)
* Building blocks or sugar cubes
* Plate
* Table

STEPS:

1. Make jello in a flat pan and let set
2. Build structures on top of jello
3. Gently shake the pan
4. Observe which buildings fall first
5. Redesign buildings to withstand shaking

THE SCIENCE BEHIND IT:

Earthquakes happen when tectonic plates move suddenly. Soft ground (like jello) amplifies shaking. Wider, lower buildings are more stable.

WHAT DID YOU OBSERVE?

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EXPERIMENT #33

Cloud in a Bottle

Difficulty: Medium   Time: 10 min

MATERIALS:

* Clear plastic bottle
* Warm water
* Match (adult help)
* Flashlight

STEPS:

1. Add small amount of warm water to bottle
2. Squeeze and release bottle a few times
3. Have adult light match and drop in bottle
4. Squeeze bottle hard then release quickly
5. Shine flashlight to see cloud form

THE SCIENCE BEHIND IT:

Clouds form when water vapor condenses on particles. Squeezing changes pressure and temperature, causing condensation on smoke particles.

WHAT DID YOU OBSERVE?

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EXPERIMENT #34

Water Cycle in a Bag

Difficulty: Easy		Time: 2 days

MATERIALS:

* Ziplock bag
* Water
* Blue food coloring
* Tape
* Sunny window

STEPS:

1. Add water with blue food coloring to bag
2. Seal bag tightly
3. Tape to sunny window
4. Observe over 1-2 days
5. See evaporation and condensation!

THE SCIENCE BEHIND IT:

The sun heats water causing evaporation. Water vapor rises, cools on the bag’s surface, condenses into droplets, and drips back down like rain.

WHAT DID YOU OBSERVE?

EXPERIMENT #35

Soil Layers Jar

Difficulty: Easy   Time: 24 hours

MATERIALS:

* Clear jar with lid
* Garden soil
* Water
* Spoon

STEPS:

1. Fill jar 1/3 with soil
2. Add water to nearly full
3. Put lid on and shake vigorously
4. Let sit undisturbed for 24 hours
5. Observe layers: gravel, sand, silt, clay

THE SCIENCE BEHIND IT:

Different soil particles have different sizes and weights. Heavy particles (gravel, sand) settle first. Light particles (silt, clay) settle last.

WHAT DID YOU OBSERVE?

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EXPERIMENT #36

Fossil Making

Difficulty: Easy Time: 24 hours

MATERIALS:

* Clay or plaster of paris
* Shells, leaves, toy dinosaurs
* Petroleum jelly
* Container
* Water

STEPS:

1. Coat objects in petroleum jelly
2. Press objects into flattened clay
3. Remove objects carefully
4. Let clay dry 24 hours
5. Observe your fossil impressions!

THE SCIENCE BEHIND IT:

Real fossils form when organisms are buried in sediment. Over millions of years, minerals replace organic material, preserving the shape.

WHAT DID YOU OBSERVE?

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EXPERIMENT #37

Wind Direction Finder

Difficulty: Easy		Time: 20 min

MATERIALS:

* Straw
* Pin
* Pencil with eraser
* Construction paper
* Compass

STEPS:

1. Cut arrow point and tail from paper
2. Tape to straw ends (tail bigger)
3. Push pin through straw center into eraser
4. Find north with compass
5. Place outside and observe wind direction

THE SCIENCE BEHIND IT:

Wind vanes point INTO the wind. The larger tail catches wind and pushes away, making the arrow point toward the wind source direction.

WHAT DID YOU OBSERVE?

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EXPERIMENT #38

Mini Water Filter

Difficulty: Medium

Time: 25 min

MATERIALS:

* Plastic bottle (cut in half)
* Sand
* Gravel
* Cotton balls
* Dirty water
* Cup

STEPS:

1. Place cotton in bottle neck (upside down)
2. Add layer of sand
3. Add layer of gravel
4. Pour dirty water through slowly
5. Collect filtered water in cup below

THE SCIENCE BEHIND IT:

Filtration removes particles by size. Gravel catches large debris, sand catches smaller particles, and cotton catches finest particles.

WHAT DID YOU OBSERVE?

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EXPERIMENT #39

Erosion Experiment

Difficulty: Easy   Time: 15 min

MATERIALS:

* 3 trays or pans
* Soil
* Grass or leaves
* Rocks
* Watering can

STEPS:

1. Fill trays with soil (bare, grass-covered, rocky)
2. Tilt all trays at same angle
3. Pour equal water on each
4. Observe which erodes most
5. Measure soil in runoff water

THE SCIENCE BEHIND IT:

Plant roots hold soil in place. Bare soil erodes fastest. This is why deforestation leads to landslides and why farmers plant cover crops.

WHAT DID YOU OBSERVE?

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EXPERIMENT #40

Weather Station

Difficulty: Medium Time: 30 min

MATERIALS:

* Jar
* Balloon
* Straw
* Tape
* Ruler
* Notebook

STEPS:

1. Cut balloon and stretch over jar opening
2. Tape straw horizontally on balloon surface
3. Mark a scale on paper behind straw
4. Record straw position daily
5. Compare to weather reports

THE SCIENCE BEHIND IT:

This is a simple barometer. Air pressure pushes on the balloon. High pressure pushes it down (straw up). Low pressure lets it rise (straw down).

WHAT DID YOU OBSERVE?

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ENGINEERING EXPERIMENTS

Experiments 41-50

EXPERIMENT #41

Bridge Building Challenge

Difficulty: Medium Time: 30 min

MATERIALS:

* Popsicle sticks
* White glue
* String
* Small weights
* Cups

STEPS:

1. Design a bridge to span 12 inches
2. Build using only sticks and glue
3. Let glue dry completely
4. Place bridge across two cups
5. Add weights until it breaks - record max

THE SCIENCE BEHIND IT:

Triangles are the strongest shape in engineering. Distributing weight across multiple points and using tension members increases bridge strength.

WHAT DID YOU OBSERVE?

EXPERIMENT #42

Balloon Rocket Car

Difficulty: Easy Time: 15 min

MATERIALS:

* Balloon
* Straw
* Tape
* Cardboard
* Bottle caps
* Skewers

STEPS:

1. Cut cardboard rectangle for car body
2. Push skewers through for axles
3. Attach bottle cap wheels
4. Tape straw on top, insert balloon end
5. Blow up balloon and release to race!

THE SCIENCE BEHIND IT:

Newton’s Third Law: every action has an equal opposite reaction. Air rushing backward out of the balloon pushes the car forward.

WHAT DID YOU OBSERVE?

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EXPERIMENT #43

Spaghetti Tower

Difficulty: Easy

Time: 20 min

MATERIALS:

* Dry spaghetti
* Marshmallows
* Tape measure
* Timer

STEPS:

1. Use marshmallows to connect spaghetti
2. Build the tallest tower possible in 15 min
3. Tower must stand on its own
4. Measure height when time is up
5. What design worked best?

THE SCIENCE BEHIND IT:

Tall structures need wide bases for stability. Triangular bracing prevents buckling. The center of gravity must stay over the base.

WHAT DID YOU OBSERVE?

EXPERIMENT #44

Rubber Band Helicopter

Difficulty: Medium   Time: 20 min

MATERIALS:

* Paper
* Rubber band
* Paper clip
* Scissors
* Pencil

STEPS:

1. Cut paper strip 1x8 inches
2. Fold in half and bend ends opposite ways
3. Attach paper clip at fold
4. Hook rubber band to paper clip
5. Wind rubber band and release!

THE SCIENCE BEHIND IT:

The twisted rubber band stores elastic potential energy. When released, it spins the paper blades. Angled blades push air down, creating lift.

WHAT DID YOU OBSERVE?

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EXPERIMENT #45

Water Wheel Generator

Difficulty: Medium

Time: 25 min

MATERIALS:

* Plastic cups (small)
* Cardboard circle
* Pencil or dowel
* Water jug
* Tape
* String

STEPS:

1. Tape small cups around cardboard circle edge
2. Push pencil through center as axle
3. Tie string to axle with small weight
4. Hold under running water
5. Watch wheel spin and lift the weight!

THE SCIENCE BEHIND IT:

Flowing water has kinetic energy. The wheel converts this into rotational energy. Real hydroelectric dams use this principle to generate electricity.

WHAT DID YOU OBSERVE?

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EXPERIMENT #46

Newspaper Table

Difficulty: Easy		Time: 15 min

MATERIALS:

* Newspaper
* Tape
* Scissors
* Heavy book for testing

STEPS:

1. Roll newspaper sheets into tight tubes
2. Tape tubes securely
3. Build a table structure with tube legs
4. Add flat newspaper top
5. Test how much weight it holds

THE SCIENCE BEHIND IT:

Tubes are much stronger than flat sheets because they resist bending in all directions. This is why bones and pipes are cylindrical.

WHAT DID YOU OBSERVE?

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EXPERIMENT #47

Zip Line Engineering

Difficulty: Easy Time: 15 min

MATERIALS:

* String or fishing line
* Straw
* Tape
* Small toy figure
* Two high/low points

STEPS:

1. Tie string between two points (angled)
2. Thread straw onto string
3. Tape toy figure to straw
4. Release from high end
5. Observe speed and adjust angle

THE SCIENCE BEHIND IT:

Gravity pulls objects downhill along the line. Steeper angles mean more speed. Friction between straw and string slows the rider.

WHAT DID YOU OBSERVE?

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EXPERIMENT #48

Parachute Design

Difficulty: Easy　　Time: 20 min

MATERIALS:

* Plastic bags
* String
* Small toy
* Scissors
* Tape
* Stopwatch

STEPS:

1. Cut squares of different sizes from bags
2. Attach 4 equal strings to corners
3. Tie strings to small toy
4. Drop from same height
5. Time each - which falls slowest?

THE SCIENCE BEHIND IT:

Parachutes work by increasing air resistance (drag). Larger surface area catches more air, creating more drag and slower descent.

WHAT DID YOU OBSERVE?

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EXPERIMENT #49

Marble Run Builder

Difficulty: Medium Time: 30 min

MATERIALS:

* Cardboard tubes
* Tape
* Scissors
* Box
* Marble

STEPS:

1. Cut tubes in half lengthwise for tracks
2. Tape tracks inside box at angles
3. Create turns and drops
4. Test marble from top
5. Redesign until marble reaches bottom

THE SCIENCE BEHIND IT:

Gravity accelerates the marble downhill. Angles and curves change the direction of momentum. Friction slows it on flat sections.

WHAT DID YOU OBSERVE?

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EXPERIMENT #50

Straw Rocket Launcher

Difficulty: Easy　　Time: 15 min

MATERIALS:

* Paper
* Straws (2 sizes)
* Tape
* Scissors
* Clay

STEPS:

1. Roll paper tightly around larger straw
2. Tape paper tube and seal one end with clay
3. Slide paper rocket over smaller straw
4. Blow hard into smaller straw
5. Measure how far rocket flies!

THE SCIENCE BEHIND IT:

Your breath creates air pressure inside the straw. This pressure pushes the sealed rocket off the end. The clay adds mass for stability.

WHAT DID YOU OBSERVE?

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____

MY EXPERIMENT JOURNAL

Experiment Name: ____

Date: ________________ Hypothesis: _____________________________

What happened: ____

____

Was my hypothesis correct? YES / NO

What I learned: ____

____

Draw your results:

MY EXPERIMENT JOURNAL

Experiment Name: ____

Date: ________________ Hypothesis: ____________________________

What happened: ____

____

Was my hypothesis correct? YES / NO

What I learned: ____

____

Draw your results:

MY EXPERIMENT JOURNAL

Experiment Name: ____

Date: ________________ Hypothesis: ____________________________

What happened: ____

____

Was my hypothesis correct? YES / NO

What I learned: ____

____

Draw your results:

YOUNG SCIENTIST CERTIFICATE

Awarded to:

____

For completing this entire book!

Date: __________________

Author: Daniel Tesfamariam

MY SCIENCE JOURNAL - Entry 1

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY SCIENCE JOURNAL - Entry 2

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY SCIENCE JOURNAL - Entry 3

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY SCIENCE JOURNAL - Entry 4

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

MY SCIENCE JOURNAL - Entry 5

Date: ________________

Today I learned:

How I will use this:

My goal for tomorrow:

Free writing space:

SCIENCE ACTIVITY #1

CHALLENGE:

Think of 3 new ideas related to this topic and explain each one.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

SCIENCE ACTIVITY #2

CHALLENGE:

Create a mind map connecting everything you learned.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

SCIENCE ACTIVITY #3

CHALLENGE:

Write a letter explaining this topic to a younger child.

My Work:

Self-rating (circle): AMAZING / GOOD / NEEDS MORE WORK

CREATIVE SPACE: SCIENCE

Draw what you learned today

Use this space for drawing, writing, or brainstorming!

CREATIVE SPACE: SCIENCE

Illustrate your favorite concept

Use this space for drawing, writing, or brainstorming!

NOTES & IDEAS

NOTES & IDEAS

NOTES & IDEAS

NOTES & IDEAS

NOTES & IDEAS